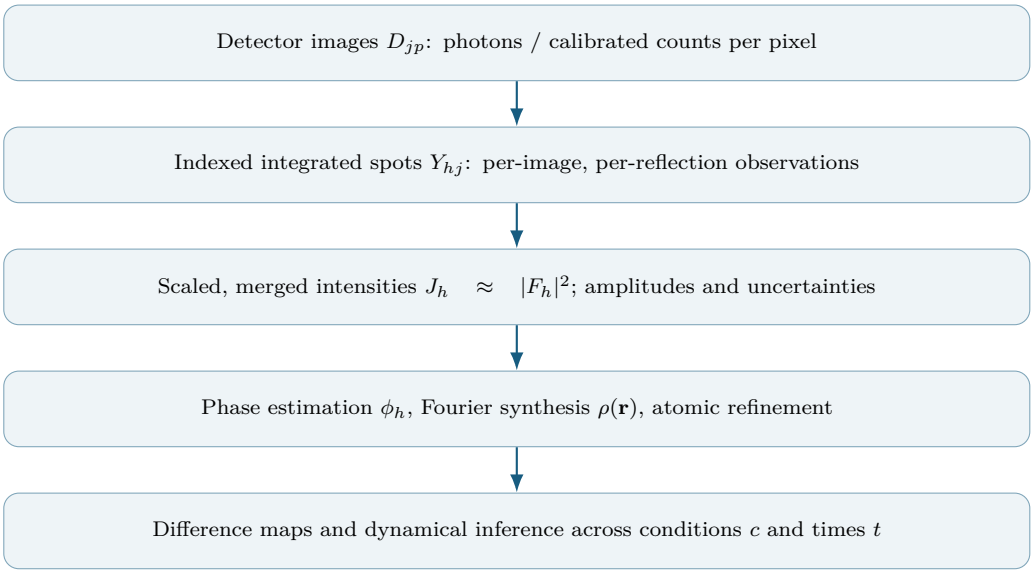
The Mathematics of X-ray Crystallography

From detector photons to electron density, atomic models, difference maps, and protein mechanics

A detailed derivation-based study companion

With special sections on the Doeke Hekstra laboratory,
EF-X, comparative crystallography, Careless, ReciprocalSpaceship,
MatchMaps, and mechanically perturbed PP2A / DNA springs

The central inversion. A crystal's electron density has complex-valued Fourier coefficients $F_{hkl} = |F_{hkl}|e^{i\phi_{hkl}}$. The detector primarily measures diffraction intensities proportional to $|F_{hkl}|^2$, *not* the phases ϕ_{hkl}. Processing converts pixel counts into reflection observations, infers their common amplitudes, supplies or estimates phases, and Fourier-synthesizes an electron-density map. Structural refinement then adjusts atomic coordinates and disorder parameters against the data with chemical restraints.



Prepared October 2026 | Technical study edition with source notes and derivations

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1 Scope, notation, and the exact inverse problem

This document emphasizes **downstream computational analysis**: spot finding, indexing, integration, symmetry assignment, scaling, merging, phasing, electron-density reconstruction, structural refinement, validation, and comparative or time-resolved inference. The experimental physics is developed only as far as necessary to derive the computational equations. It is designed to complement a similarly mathematical treatment of single-particle cryo-EM.

Inputs and outputs. A diffraction image is a 2D pixel array. An indexed reflection is a reciprocal-lattice label (h, k, l), not a spatial point in the protein. A merged reflection table contains at least a label, intensity or amplitude, and uncertainty. A density map is a 3D function over a periodic unit cell. An atomic model is a list of atom positions, occupancies and displacement parameters. These are *different mathematical types*; a software workflow transforms one into another.

Symbol	Precise meaning
$\mathbf{r} \in \mathbb{R}^3$, $\mathbf{x} \in [0,1)^3$	Cartesian and fractional real-space coordinates, respectively.
$\rho(\mathbf{r})$	Electron density, measured in electrons per cubic Å in the idealized interpretation.
$\mathbf{a}, \mathbf{b}, \mathbf{c}$; V_c	Unit-cell basis vectors and unit-cell volume, $V_c = \lvert\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})\rvert$.
$\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*$	Reciprocal basis vectors in cycles per length (no 2π in their definition).
$\mathbf{h} = (h, k, l) \in \mathbb{Z}^3$	Miller-index triplet labeling a reciprocal-lattice point.
$F_{\mathbf{h}}$, $\phi_{\mathbf{h}}$	Complex structure factor and its phase: $F_{\mathbf{h}} = \lvert F_{\mathbf{h}}\rvert e^{i\phi_{\mathbf{h}}}$.
j, p	Detector-image/frame index and pixel index within an image.
$Y_{\mathbf{h}j}$, $\sigma_{\mathbf{h}j}$	Integrated observation and its estimated standard uncertainty.
$J_{\mathbf{h}}$	Ideal merged intensity on a chosen common scale, approximately $\lvert F_{\mathbf{h}}\rvert^2$.
$\Sigma_{\mathbf{h}j}$	Positive multiplicative observation scale (care with reciprocal conventions).
$\boldsymbol{\theta}$	Parameters: atomic coordinates, B values, occupancies, scale models, etc., depending on context.
c, t	Perturbation condition and measurement time after perturbation.

2 Detector data: what the pixels actually store

2.1 Raw frames are matrices, not reflection tables

For diffraction image j, let $D_j[p]$ denote the calibrated detector count at pixel $p = (u, v)$. It may be modeled as

$$D_j[p] = B_j[p] + S_j[p] + \epsilon_j[p], \tag{1}$$

where $B_j[p]$ is background (air scatter, solvent, detector offset, etc.), $S_j[p]$ is X-ray scattering signal, and $\epsilon_j[p]$ describes noise or residual corrections. A detector may initially register photon counts with a Poisson-like distribution,

$$N_j[p] \sim \text{Poisson}(\mu_j[p]), \qquad P(N = n \mid \mu) = e^{-\mu}\frac{\mu^n}{n!}. \tag{2}$$

The Poisson parameter μ is the expected number of recorded counts; its variance is also μ. After corrections, counts need not be integers or obey an exact Poisson model. Hot pixels, gain correction, trusted ranges, saturation and masks affect pixel usability.

Input: arrays $D_j \in \mathbb{R}^{H \times W}$, detector geometry, pixel size, detector distance, beam center, wavelength, and rotation metadata. **Output:** calibrated images and a geometric model that maps pixel coordinates to scattering vectors.

2.2 A pixel corresponds to a scattered-wave direction

Let the incident wavevector be $\mathbf{k}_0$ and the scattered wavevector associated with pixel p be $\mathbf{k}_p$, both having magnitude $1/\lambda$ in the *cycles-per-length* convention. Define

$$\mathbf{q}_p = \mathbf{k}_p - \mathbf{k}_0, \qquad |\mathbf{k}_p| = |\mathbf{k}_0| = \lambda^{-1}. \tag{3}$$

The unit direction $\widehat{\mathbf{s}}_p$ from the sample to pixel p yields $\mathbf{k}_p = \widehat{\mathbf{s}}_p/\lambda$. If the detector is perpendicular to the incident beam at distance L and pixel offsets from the beam center are (X, Y), then $\widehat{\mathbf{s}}_p = (X, Y, L)/\sqrt{X^2 + Y^2 + L^2}$. Real instruments use the refined detector-panel geometry instead. The scattering angle between $\mathbf{k}_0$ and $\mathbf{k}_p$ is 2θ.

3 Direct lattice and reciprocal lattice from first principles

3.1 A periodic unit cell in real space

A crystal repeats by lattice translations

$$\mathbf{R}_n = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}, \qquad n_i \in \mathbb{Z}. \tag{4}$$

The basis matrix is $A = [\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$. Any real-space position can be written $\mathbf{r} = A\mathbf{x}$ with fractional coordinates $\mathbf{x} = (x, y, z)^{\mathsf{T}}$. Unit-cell volume is $V_c = |\det A|$; it converts a fractional-volume element into Cartesian volume: $\mathrm{d}^3\mathbf{r} = V_c\mathrm{d}^3\mathbf{x}$.

3.2 Reciprocal basis and its proof

Define $B = A^{-\mathsf{T}}$ and take its columns as $\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*$. Then

$$\mathbf{a}^* \cdot \mathbf{a} = 1, \quad \mathbf{a}^* \cdot \mathbf{b} = \mathbf{a}^* \cdot \mathbf{c} = 0, \tag{5}$$

with cyclic permutations. For example, $\mathbf{a}^* = (\mathbf{b} \times \mathbf{c})/[\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})]$. For Miller indices $\mathbf{h} = (h, k, l)^{\mathsf{T}}$ the reciprocal vector is

$$\mathbf{g}_\mathbf{h} = B\mathbf{h} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*. \tag{6}$$

The dot product with a lattice translation is $\mathbf{g}_\mathbf{h} \cdot \mathbf{R}_n = \mathbf{h}^{\mathsf{T}}\mathbf{n} \in \mathbb{Z}$, hence

$$e^{2\pi i \mathbf{g}_\mathbf{h} \cdot \mathbf{R}_n} = 1. \tag{7}$$

This identity proves that the reciprocal phase factors remain unchanged under any unit-cell translation, which is why periodic crystals scatter coherently at reciprocal-lattice points.

3.3 Resolution and the reciprocal metric tensor

The spacing of planes indexed $\mathbf{h}$ is

$$\boxed{d_\mathbf{h} = \frac{1}{|\mathbf{g}_\mathbf{h}|} = \frac{1}{\sqrt{\mathbf{h}^{\mathsf{T}} G^* \mathbf{h}}}}, \qquad G^* = B^{\mathsf{T}}B. \tag{8}$$

The reciprocal metric G^* handles any triclinic unit cell. For orthogonal lengths (a, b, c), $1/d_{hkl}^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$. Reciprocal vectors are in Å^{-1}, while d is in Å. The index (h, k, l) is dimensionless.

Worked calculation

For a cubic crystal with side $a = 50\,\text{Å}$, reflection $(3, 4, 0)$ has

$$|\mathbf{g}_{340}| = \sqrt{3^2 + 4^2}/50 = 0.10\,\text{Å}^{-1}, \qquad d_{340} = 10\,\text{Å}.$$

The Miller indices identify a *Fourier component* with a spatial period of 10 Å; they do not identify a protein atom at coordinate $(3, 4, 0)$.

3.4 Bragg's law and the Ewald construction

Elastic scattering enforces $|\mathbf{k}_p| = |\mathbf{k}_0| = 1/\lambda$. A Bragg reflection occurs when

$$\boxed{\mathbf{q} = \mathbf{g}_\mathbf{h}} \tag{9}$$

for some reciprocal-lattice point (up to the orientation mapping to laboratory coordinates). From the isosceles triangle of elastic wavevectors, $|\mathbf{q}| = 2\sin\theta/\lambda$, so $1/d = 2\sin\theta/\lambda$, giving

$$\boxed{2d_{\mathbf{h}}\sin\theta = \lambda.} \tag{10}$$

The Ewald sphere is the sphere of allowable outgoing $\mathbf{k}_p$ endpoints; crystal rotation sweeps reciprocal points through this sphere. Bragg's law supplies a diffraction condition, not an electron-density reconstruction algorithm.

4 Why a crystal makes sharp Bragg reflections

4.1 Lattice sum and interference

Suppose one unit cell has scattering amplitude $F(\mathbf{q})$, and the finite crystal contains lattice translations $\{\mathbf{R}_n\}$. The total amplitude is

$$A(\mathbf{q}) = F(\mathbf{q})\sum_n e^{2\pi i\mathbf{q}\cdot\mathbf{R}_n}. \tag{11}$$

The detector intensity is proportional to $|A(\mathbf{q})|^2$. At $\mathbf{q} = \mathbf{g}_{\mathbf{h}}$, all exponential factors equal one and the amplitudes add coherently. Away from reciprocal points, phases cancel approximately. Finite crystal size broadens peaks: a Bragg reflection has a footprint across detector pixels and often spans several rotation images rather than being an ideal delta function.

4.2 Bragg scattering is not the entire scattering signal

Averages of periodically repeated unit cells produce sharp Bragg scattering. Correlated departures from the average can produce diffuse scattering between Bragg positions. Distinguish a mean density from fluctuating density:

$$\rho_n(\mathbf{r}) = \bar{\rho}(\mathbf{r}) + \delta\rho_n(\mathbf{r}), \quad \mathbb{E}[\delta\rho_n] = 0. \tag{12}$$

Bragg peaks primarily constrain the periodic mean structure; diffuse scattering can encode covariance of fluctuations. This matters when interpreting protein dynamics: refined B factors and occupancies alone cannot supply a complete map of correlated motion.

5 The complex structure factor: Fourier transform of electron density

5.1 Derivation in fractional coordinates

Let $\rho(\mathbf{r})$ be the unit-cell electron density, periodic under lattice translations. Using fractional $\mathbf{x} = A^{-1}\mathbf{r}$, define

$$\boxed{F_{\mathbf{h}} = \int_{\text{unit cell}} \rho(\mathbf{r})e^{2\pi i\mathbf{g}_{\mathbf{h}}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{r} = V_c\int_{[0,1)^3}\rho(A\mathbf{x})e^{2\pi i\mathbf{h}\cdot\mathbf{x}}\,\mathrm{d}^3\mathbf{x}.} \tag{13}$$

The sign convention here uses $+$ in the forward transform and $-$ in the synthesis; either consistent convention is valid. $F_{\mathbf{h}}$ is complex and represents the Fourier coefficient for the reflection $\mathbf{h}$. Its units, for density in electrons per Å^3 and a volume integral, are electrons in the idealized convention.

5.2 From atomic coordinates to calculated structure factors

For M atoms in the unit cell, one common approximation is

$$\boxed{F_{\text{calc},\mathbf{h}} = \sum_{j=1}^{M} o_j f_j(s)e^{-B_j s^2/4}e^{2\pi i\mathbf{h}\cdot\mathbf{x}_j}, \qquad s = |\mathbf{g}_{\mathbf{h}}| = 1/d_{\mathbf{h}}.} \tag{14}$$

Here $o_j \in [0,1]$ is occupancy; $f_j(s)$ is the X-ray atomic scattering factor at frequency s; B_j is the isotropic atomic displacement parameter (Å^2); $\mathbf{x}_j$ is the fractional coordinate of atom j. The convention $e^{-B_j s^2/4}$ corresponds

to $B = 8\pi^2 U_{\text{iso}}$. For anisotropic displacement matrix U_j in Cartesian units, the attenuation is $\exp[-2\pi^2 \mathbf{g}^{\mathsf{T}} U_j \mathbf{g}]$. Model calculations also include symmetry, solvent and other corrections.

Every atom contributes a complex phasor. The sum is not a simple sum of atomic intensities: phases interfere. Two atoms may reinforce or cancel each other's contribution at a given reflection.

5.3 Why the phase matters: a two-atom proof

Take identical atoms at fractional positions $x_1 = 0$ and $x_2 = \frac{1}{2}$ in one dimension. Ignoring atomic form factors, for reflection integer h,

$$F_h = 1 + e^{2\pi \mathrm{i} h(1/2)} = 1 + (-1)^h. \tag{15}$$

Thus $F_h = 2$ for even h but $F_h = 0$ for odd h. Although both atoms scatter, odd-index reflections cancel exactly. This reveals an interference effect encoded in the complex structure factor.

5.4 The inverse Fourier series

The unit-cell electron density is recovered by

$$\boxed{\rho(A\mathbf{x}) = \frac{1}{V_c} \sum_{\mathbf{h} \in \mathbb{Z}^3} F_{\mathbf{h}} e^{-2\pi \mathrm{i} \mathbf{h}\cdot\mathbf{x}}.} \tag{16}$$

Proof sketch: substitute the Fourier-series expansion into Eq. (13) and use orthogonality, $\int_{[0,1)^3} e^{2\pi \mathrm{i}(\mathbf{h}-\mathbf{h}')\cdot\mathbf{x}} \mathrm{d}^3\mathbf{x} = \delta_{\mathbf{h},\mathbf{h}'}$. The integral selects exactly one coefficient. Finite resolution truncates the sum, so experimental maps are band-limited and contain termination artifacts.

6 Intensity, amplitude, and the phase problem

Write a complex structure factor in polar form:

$$F_{\mathbf{h}} = |F_{\mathbf{h}}| e^{\mathrm{i}\phi_{\mathbf{h}}}. \tag{17}$$

The measured integrated intensity is approximately

$$\boxed{J_{\mathbf{h}} \propto |F_{\mathbf{h}}|^2.} \tag{18}$$

After an appropriate scale convention, one writes $J_{\mathbf{h}} \approx |F_{\mathbf{h}}|^2$. Measurement contains the squared magnitude, not the phase $\phi_{\mathbf{h}}$. The loss of the angle in the complex plane is the *phase problem*.

Worked calculation

Two complex coefficients $F_1 = 3 + 4\mathrm{i}$ and $F_2 = 3 - 4\mathrm{i}$ both produce intensity $|F|^2 = 3^2 + 4^2 = 25$ but carry phases of opposite sign. Their individual contributions to real-space density can differ. Knowing $|F| = 5$ is insufficient to decide between these coefficients.

6.1 Translation changes phases without changing intensities

If the electron density shifts by fractional vector $\mathbf{x}_0$, $\rho'(\mathbf{x}) = \rho(\mathbf{x} - \mathbf{x}_0)$, the Fourier coefficient becomes

$$F'_{\mathbf{h}} = e^{2\pi \mathrm{i} \mathbf{h}\cdot\mathbf{x}_0} F_{\mathbf{h}}, \qquad |F'_{\mathbf{h}}|^2 = |F_{\mathbf{h}}|^2. \tag{19}$$

The equality follows by changing variables in the Fourier integral and using $|e^{\mathrm{i}\alpha}| = 1$. Intensities alone cannot determine a unique absolute origin. More generally, distinct density arrangements can share some or all intensity data: the phase problem is genuinely underdetermined without additional constraints or information.

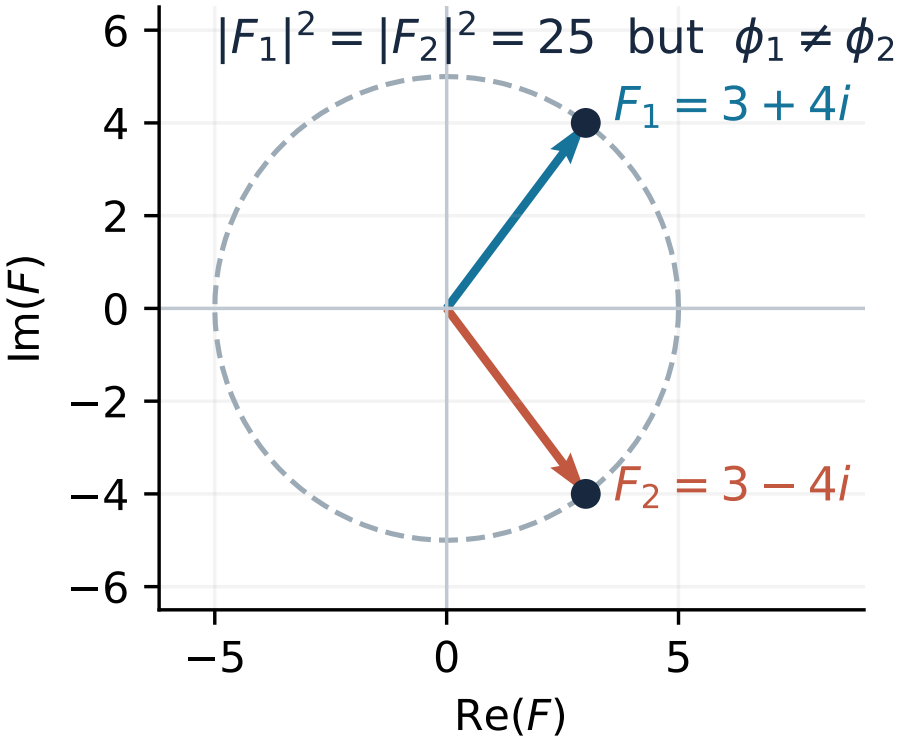


Figure 1: Two different complex structure factors of equal magnitude: an intensity measurement cannot distinguish their phases. This diagram is a mathematical example, not experimental data.

7 Indexing: from peak coordinates to Miller indices

7.1 The orientation matrix

After peak detection, the program knows scattering vectors $\mathbf{q}_p$ from detector geometry, but not which (h, k, l) produced each spot. Let $U \in SO(3)$ rotate the crystal-frame reciprocal lattice into the laboratory frame and $B = A^{-\mathsf{T}}$. Then

$$\boxed{\mathbf{q}_\mathbf{h} = UB\mathbf{h}.} \tag{20}$$

For a rotation scan at spindle angle ω_j, an additional laboratory rotation R_{ω_j} gives $\mathbf{q}_{\mathbf{h},j} = R_{\omega_j} UB\mathbf{h}$. Indexing estimates lattice metric and orientation from many peak locations; the integer vector $\mathbf{h}$ is assigned so that the predicted reciprocal vectors agree with measured ones.

7.2 A least-squares formulation

With measured reciprocal vectors $\widetilde{\mathbf{q}}_r$ for candidate spots r, a simplified indexing objective is

$$\min_{U,B,\{\mathbf{h}_r \in \mathbb{Z}^3\}} \sum_r w_r \|\widetilde{\mathbf{q}}_r - R_{\omega_r} UB\mathbf{h}_r\|^2, \tag{21}$$

subject to unit-cell constraints and valid lattice symmetry. w_r reflects peak-position uncertainty. This is a mixed continuous/discrete fitting problem; practical DIALS/XDS algorithms use specialized lattice searches and geometry refinement rather than brute-force minimization.

Input: detected centroids, crystal rotation angles, wavelength and detector geometry. **Output:** unit-cell parameters, crystal orientation, indexed Miller labels, and predicted reflection positions. Unindexed peaks may reflect artifacts, additional lattices, or indexing failure.

8 Integration: how a spot becomes a measured intensity

8.1 Summing calibrated pixels after background subtraction

A Bragg peak occupies a spatial footprint over detector pixels and sometimes adjacent rotation frames. If $\mathcal{P}_{\mathbf{h}j}$ is a pixel region associated with reflection $\mathbf{h}$ on image j, a basic sum estimate is

$$\boxed{Y_{\mathbf{h}j} = \sum_{p \in \mathcal{P}_{\mathbf{h}j}} [D_j[p] - \widehat{B}_j[p]].} \tag{22}$$

$\widehat{B}_j[p]$ is an estimated local background and $Y_{\mathbf{h}j}$ is the integrated reflection observation. Note that it can be *negative*: background subtraction and noise can exceed a weak true spot signal. This does not imply a physically negative scattering intensity.

If the corrected counts are independent and variances are known, a first-order uncertainty estimate is

$$\mathrm{Var}(Y_{\mathbf{h}j}) \approx \sum_{p \in \mathcal{P}_{\mathbf{h}j}} [\mathrm{Var}(D_j[p]) + \mathrm{Var}(\widehat{B}_j[p])], \tag{23}$$

with covariance terms needed when background estimates share data or pixel noise is correlated. The standard uncertainty is $\sigma_{\mathbf{h}j} = \sqrt{\mathrm{Var}(Y_{\mathbf{h}j})}$.

8.2 Profile fitting as weighted least squares

A more efficient estimate at low SNR uses an expected pixel profile P_p for a reflection. Consider the model $D_p = B_p + IP_p + \epsilon_p$ with $\sum_p P_p = 1$, where I is the unknown integrated intensity. Under independent Gaussian errors with variances σ_p^2, minimize

$$L(I) = \sum_p \frac{(D_p - B_p - IP_p)^2}{\sigma_p^2}. \tag{24}$$

Differentiating and setting to zero yields

$$\boxed{\widehat{I} = \frac{\sum_p P_p(D_p - B_p)/\sigma_p^2}{\sum_p P_p^2/\sigma_p^2}}, \qquad \mathrm{Var}(\widehat{I}) = \frac{1}{\sum_p P_p^2/\sigma_p^2}. \tag{25}$$

This derivation assumes the fitted profile and background are known and the stated noise model is correct. Actual integration models also account for spot overlap, partiality, changing reflection geometry, masks and detector effects.

Worked calculation

Suppose three background-corrected pixels are $y = (7, 12, 6)$ counts and a normalized spot profile is $P = (0.25, 0.50, 0.25)$. With equal variance $\sigma_p^2 = 4$, the fit gives $\widehat{I} = (0.25 \cdot 7 + 0.50 \cdot 12 + 0.25 \cdot 6)/(0.25^2 + 0.50^2 + 0.25^2) = 9.25/0.375 \approx 24.67$ counts, while the standard uncertainty is $2/\sqrt{0.375} \approx 3.27$ counts. The fitted integrated intensity is not the central pixel's value.

9 Symmetry, equivalent reflections, and systematic absences

9.1 Point symmetry versus translational symmetry

A space-group operation in fractional coordinates is

$$\mathbf{x}' = W\mathbf{x} + \mathbf{t} \pmod 1, \tag{26}$$

where W is a crystallographic integer rotation matrix and $\mathbf{t}$ a fractional translation. Under crystallographic symmetry the density is invariant: $\rho(W\mathbf{x} + \mathbf{t}) = \rho(\mathbf{x})$.

Substituting into the structure-factor integral gives the reciprocal-space relation

$$\boxed{F_{\mathbf{h}} = e^{2\pi i \mathbf{h}\cdot\mathbf{t}} F_{W^{\intercal}\mathbf{h}},} \tag{27}$$

up to equivalent algebraic conventions for how the symmetry operation is applied. Since the leading factor has unit magnitude, symmetry-related reflections have equal ideal intensities (ignoring anomalous and experimental effects). Translation components create predictable phase relationships; in some cases they imply $F_{\mathbf{h}} = 0$, giving systematic absences used for space-group determination.

9.2 Friedel's law and anomalous exceptions

For a real electron density (neglecting anomalous dispersion),

$$F_{-\mathbf{h}} = F_{\mathbf{h}}^{*}, \qquad |F_{-\mathbf{h}}| = |F_{\mathbf{h}}|. \tag{28}$$

This is Friedel's law and follows immediately by complex conjugation of Eq. (13). With anomalous scattering, $f_j = f_j' + \mathrm{i}f_j''$ becomes complex; the equality can fail, so Friedel mates may have different intensities, providing phasing information.

Input: indexed reflections and symmetry tests. **Output:** a consistent symmetry interpretation, asymmetric-unit reflection set, multiplicities, and an understanding of which observations should share an underlying intensity.

10 Scaling: correcting systematic variation between observations

10.1 Why identical reflections do not have identical measured intensities

The reflection $\mathbf{h}$ may be measured many times across images and crystals. Differences in illumination, radiation damage, absorption, detector response, crystal size, partiality and geometry make the observations unequal. A useful *forward model* is

$$\boxed{Y_{\mathbf{h}j} = \Sigma_{\mathbf{h}j} J_{\mathbf{h}} + \epsilon_{\mathbf{h}j}, \qquad \Sigma_{\mathbf{h}j} > 0, \quad J_{\mathbf{h}} = |F_{\mathbf{h}}|^2 \geq 0.} \tag{29}$$

Here Σ is a multiplicative factor from true intensity to measured intensity. Some crystallographic software instead defines a *correction* $K = 1/\Sigma$ so that KY is the corrected observation; always check a program's convention. ϵ is random observation error.

A classical parametric form may combine a scale s_j, a resolution-dependent attenuation and other terms, e.g.,

$$\Sigma_{\mathbf{h}j} = s_j \exp[-\tfrac{1}{2}\mathbf{g}_{\mathbf{h}}^{\intercal} B_j^{\mathrm{aniso}} \mathbf{g}_{\mathbf{h}}]\, P_{\mathbf{h}j}, \tag{30}$$

where B_j^{aniso} is a convention-dependent anisotropic scale tensor and P is a partiality/geometry correction term. The precise parameterization differs by software and experiment. It is important not to confuse an overall diffraction scale B_j^{aniso} with an individual atom's displacement parameter B_j.

10.2 Weighted merging is an estimator, not raw addition

Assume $\Sigma_{\mathbf{h}j}$ is known and observations are independent Gaussian variables with variances $\sigma_{\mathbf{h}j}^2$. The negative log likelihood for one reflection, up to constants, is

$$L(J_{\mathbf{h}}) = \frac{1}{2} \sum_j \frac{(Y_{\mathbf{h}j} - \Sigma_{\mathbf{h}j} J_{\mathbf{h}})^2}{\sigma_{\mathbf{h}j}^2}. \tag{31}$$

Differentiate:

$$\frac{\mathrm{d}L}{\mathrm{d}J_{\mathbf{h}}} = -\sum_j \frac{\Sigma_{\mathbf{h}j}(Y_{\mathbf{h}j} - \Sigma_{\mathbf{h}j} J_{\mathbf{h}})}{\sigma_{\mathbf{h}j}^2}. \tag{32}$$

Set derivative to zero, rearrange, and obtain

$$\boxed{\widehat{J}_{\mathbf{h}} = \frac{\sum_j \Sigma_{\mathbf{h}j} Y_{\mathbf{h}j} / \sigma_{\mathbf{h}j}^2}{\sum_j \Sigma_{\mathbf{h}j}^2 / \sigma_{\mathbf{h}j}^2}}, \qquad \boxed{\mathrm{Var}(\widehat{J}_{\mathbf{h}}) = \left(\sum_j \frac{\Sigma_{\mathbf{h}j}^2}{\sigma_{\mathbf{h}j}^2} \right)^{-1}}. \tag{33}$$

These are maximum-likelihood estimates under the stated assumptions. This is equivalent to a variance-weighted average of scaled intensities Y_j/Σ_j, with weights Σ_j^2/σ_j^2. In reality the scales must also be inferred, creating a coupled estimation problem.

Worked calculation

Take a single reflection measured twice, with $Y_1 = 120, \sigma_1 = 10, \Sigma_1 = 2$ and $Y_2 = 48, \sigma_2 = 8, \Sigma_2 = 1$. The corrected values are 60 ± 5 and 48 ± 8. Their inverse-variance weights are $1/25$ and $1/64$. The merged estimate is

$$\widehat{J} = \frac{60/25 + 48/64}{1/25 + 1/64} = 56.63, \qquad \sigma_{\widehat{J}} = \frac{1}{\sqrt{1/25 + 1/64}} \approx 4.24.$$

A raw arithmetic average of 120 and 48 would be meaningless because they were measured on different scales.

10.3 An identifiability issue: global scale ambiguity

If the forward prediction is $Y_{\mathbf{h}j} = \Sigma_{\mathbf{h}j} J_{\mathbf{h}}$, replacing $\Sigma \to a\Sigma$ and $J \to J/a$ for any positive a leaves every prediction unchanged. Therefore the absolute scale cannot be inferred from data alone without a gauge choice, normalization or prior. This is an example of a *non-identifiable* direction in parameter space.

11 Reflection-data quality statistics and what they mean

11.1 Completeness, multiplicity, and signal-to-noise

Completeness is #{unique measured reflections}/#{possible unique reflections} within a resolution shell and symmetry interpretation. Multiplicity is the number of observations per unique reflection, averaged over a set. I/σ_I is an observation's or merged reflection's ratio of estimated intensity to its standard uncertainty; it is not a probability and is not universally Gaussian at low count rates.

11.2 R_{merge} and its dependence on redundancy

For scaled equivalent observations $I_{\mathbf{h}j}$ and their mean $\bar{I}_{\mathbf{h}}$,

$$R_{\text{merge}} = \frac{\sum_{\mathbf{h}} \sum_j |I_{\mathbf{h}j} - \bar{I}_{\mathbf{h}}|}{\sum_{\mathbf{h}} \sum_j I_{\mathbf{h}j}}. \tag{34}$$

This traditional definition can increase with multiplicity even if experimental quality is unchanged. For reflection multiplicity $n_{\mathbf{h}}$, redundancy-independent or precision-indicating variants include

$$R_{\text{meas}} = \frac{\sum_{\mathbf{h}} \sqrt{n_{\mathbf{h}}/(n_{\mathbf{h}} - 1)} \sum_j |I_{\mathbf{h}j} - \bar{I}_{\mathbf{h}}|}{\sum_{\mathbf{h}} \sum_j I_{\mathbf{h}j}}, \tag{35}$$

$$R_{\text{pim}} = \frac{\sum_{\mathbf{h}} \sqrt{1/(n_{\mathbf{h}} - 1)} \sum_j |I_{\mathbf{h}j} - \bar{I}_{\mathbf{h}}|}{\sum_{\mathbf{h}} \sum_j I_{\mathbf{h}j}}, \tag{36}$$

where terms with $n_{\mathbf{h}} < 2$ are excluded from these summands. The formulas assume corrected observations and positive denominator conventions.

11.3 The half-dataset correlation coefficient $\text{CC}_{1/2}$

Split observations into two independent random halves and merge each reflection separately to get $I_{\mathbf{h}}^{(A)}$ and $I_{\mathbf{h}}^{(B)}$. Define

$$\boxed{\text{CC}_{1/2} = \frac{\sum_{\mathbf{h}} (I_{\mathbf{h}}^{(A)} - \bar{I}^{(A)})(I_{\mathbf{h}}^{(B)} - \bar{I}^{(B)})}{\sqrt{\sum_{\mathbf{h}} (I_{\mathbf{h}}^{(A)} - \bar{I}^{(A)})^2 \sum_{\mathbf{h}} (I_{\mathbf{h}}^{(B)} - \bar{I}^{(B)})^2}}.} \tag{37}$$

This is a Pearson correlation across reflections, usually computed by resolution shell. A positive $CC_{1/2}$ can reveal useful signal even when mean I/σ is weak. Its interpretation depends on dataset size, correlations and the split procedure. Crystal-to-crystal generalization is not established simply by splitting individual measurements at random.

12 Bayesian amplitude inference: French–Wilson and Wilson statistics

12.1 Why negative integrated intensities must not be square-rooted

The measured, background-corrected intensity Y can be negative. A physically true intensity $J = |F|^2$ must be nonnegative. Directly setting $|F| = \sqrt{Y}$ fails for $Y < 0$; discarding all such reflections biases weak intensities upward.

The French–Wilson approach combines a measurement likelihood with a prior distribution for true intensity:

$$\boxed{p(J \mid Y, \sigma) \propto p(Y \mid J, \sigma) p(J), \qquad J \geq 0.} \tag{38}$$

For a Gaussian intensity likelihood, $p(Y \mid J, \sigma) = \mathcal{N}(Y; J, \sigma^2)$. The posterior mean amplitude is

$$\mathbb{E}[F \mid Y] = \frac{\int_0^\infty \sqrt{J}\, p(Y \mid J, \sigma) p(J)\, \mathrm{d}J}{\int_0^\infty p(Y \mid J, \sigma) p(J)\, \mathrm{d}J}. \tag{39}$$

This is generally not the same as $\sqrt{\mathbb{E}[J \mid Y]}$; nonlinear transformation and averaging do not commute.

12.2 Where Wilson's prior comes from

For many random atomic positions, the real and imaginary components of acentric structure factors are approximately Gaussian by the central limit theorem. If each is independent with variance $\Sigma_F/2$, then $J = |F|^2$ follows an exponential distribution:

$$\boxed{p(J \mid \Sigma_F) = \Sigma_F^{-1} e^{-J/\Sigma_F}, \quad J \geq 0 \qquad \text{(acentric)}.} \tag{40}$$

$\Sigma_F = \mathbb{E}[J]$ is the Wilson expected intensity in a suitable resolution and symmetry class. For a centric reflection, idealized real F is Gaussian, making $J = F^2$ chi-squared-like:

$$p(J \mid \Sigma_F) = \frac{1}{\sqrt{2\pi \Sigma_F J}} \exp\left(-\frac{J}{2\Sigma_F}\right), \quad J > 0 \quad \text{(centric)}. \tag{41}$$

Acentric and centric reflections thus require different priors. Wilson statistics are approximate and must be used with correct normalization and symmetry considerations.

13 Careless: Bayesian scaling and merging in the Hekstra lab

13.1 The motivation and the model

Hekstra-lab work by Dalton, Greisman and Hekstra developed *Careless* as a Bayesian framework that jointly models systematic scales and reflection amplitudes, rather than treating scaling, merging and French–Wilson corrections as wholly separate steps [3]. In a simplified monochromatic model,

$$\boxed{Y_{\mathbf{h}j} \mid F_{\mathbf{h}}, \Sigma_{\mathbf{h}j} \sim \mathcal{N}(\Sigma_{\mathbf{h}j} F_{\mathbf{h}}^2, \sigma_{\mathbf{h}j}^2), \quad \Sigma_{\mathbf{h}j} = s_{\boldsymbol{\theta}}(\mathbf{m}_{\mathbf{h}j}),} \tag{42}$$

where $\mathbf{m}_{\mathbf{h}j}$ contains reflection metadata (image/frame, location, resolution, geometry, etc.), and $s_{\boldsymbol{\theta}}$ is a positive scale predictor, often parameterized by a neural network. A robust Student-t likelihood is another supported option. The latent amplitude $F_{\mathbf{h}} > 0$ is endowed with a centricity-aware prior based on Wilson statistics. This is a conceptual abstraction of a flexible implementation, not an exact specification of every contemporary option.

13.2 Posterior factorization

The joint density has the structure

$$p(Y, F, \Sigma) = p(Y \mid F, \Sigma)\, p(F)\, p(\Sigma), \tag{43}$$

where $p(\Sigma)$ may represent explicit assumptions or the deterministic/stochastic scale model. Bayes' theorem gives

$$p(F, \Sigma \mid Y) \propto p(Y \mid F, \Sigma) p(F) p(\Sigma). \tag{44}$$

The exact posterior is intractable in realistic datasets. The method introduces a variational approximation $q_\varphi(F, \Sigma)$ with parameters φ.

13.3 Deriving the variational evidence lower bound

Insert the approximate posterior into the marginal likelihood:

$$\log p(Y) = \log \int q_\varphi(F, \Sigma) \frac{p(Y, F, \Sigma)}{q_\varphi(F, \Sigma)} \, \mathrm{d}F \, \mathrm{d}\Sigma \tag{45}$$

$$= \log \mathbb{E}_{q_\varphi} \left[\frac{p(Y, F, \Sigma)}{q_\varphi(F, \Sigma)} \right] \tag{46}$$

$$\geq \mathbb{E}_{q_\varphi}[\log p(Y \mid F, \Sigma)] - D_{\mathrm{KL}}(q_\varphi(F, \Sigma) \| p(F, \Sigma)). \tag{47}$$

The inequality is Jensen's inequality for the concave logarithm. Therefore

$$\boxed{\mathcal{E}(\varphi, \boldsymbol{\theta}) = \mathbb{E}_{q_\varphi}[\log p(Y \mid F, \Sigma)] - D_{\mathrm{KL}}(q_\varphi \| p_0)} \tag{48}$$

is an evidence lower bound under an appropriate prior p_0. Optimization maximizes $\mathcal{E}$ (or minimizes its negative), balancing fit to observed reflection intensities against regularization from prior assumptions. Modern implementations may parameterize parts of the scale model as learned distributions and optimize model and variational parameters jointly.

13.4 A subtle statistical point: $\mathbb{E}[F^2] \neq \mathbb{E}[F]^2$

Predicted intensity uses the *second moment* of amplitude. Under a factorized posterior with scale and amplitude independent,

$$\mathbb{E}[\widehat{Y}_{\mathbf{h}j}] = \mathbb{E}[\Sigma_{\mathbf{h}j}]\mathbb{E}[F_{\mathbf{h}}^2] = \mathbb{E}[\Sigma_{\mathbf{h}j}]\big(\mathbb{E}[F_{\mathbf{h}}]^2 + \mathrm{Var}(F_{\mathbf{h}})\big). \tag{49}$$

This is important for weak diffraction. Simply squaring the posterior mean amplitude loses the uncertainty contribution $\mathrm{Var}(F)$. The prediction variance similarly involves fourth moments $\mathbb{E}[F^4]$ and the scale's second moment. These moments can be evaluated analytically for some variational families or by sampling.

13.5 Laue / polychromatic data and harmonic overlap

For polychromatic illumination, multiple wavelength-indexed reflections may contribute to one detector spot (a ray). Then the observation model becomes

$$\boxed{Y_{rj} \sim p\left(Y_{rj} \,\middle|\, \sum_{\mathbf{h} \in \mathcal{H}(r,j)} \Sigma_{\mathbf{h}j} F_{\mathbf{h}}^2, \sigma_{rj} \right).} \tag{50}$$

The set $\mathcal{H}(r, j)$ contains contributions geometrically superposed at detector ray r and image j. The likelihood must account for *the sum of intensities*, not a simple one-to-one spot-to-reflection assignment. Careless was designed to accommodate this structure, including time-resolved Laue examples [3].

Input to Careless: integrated, unmerged observations plus uncertainties, indices and experimental metadata. **Output:** posterior estimates for reflection amplitudes and scales, uncertainty-aware predictions, and validation summaries. **Not its principal output:** atomic coordinates or phases. Those remain downstream problems.

14 ReciprocalSpaceship: reflection tables as mathematical data objects

A post-integration dataset can be organized as rows indexed by $\mathbf{h}$ (and potentially measurement identity j) and typed columns such as I, σ_I, F, σ_F, phase ϕ, or anomalous counterparts. *ReciprocalSpaceship*, developed by Greisman, Dalton and Hekstra, provides crystallography-aware tabular operations with symmetry, unit-cell and MTZ support [2].

The distinction between a table row and the physical object is crucial: a row (h, k, l, F, ϕ) refers to a Fourier coefficient, while a row $(\text{atom}, x, y, z, B, o)$ refers to an atomic model. The former describes reciprocal-space measurements or estimates; the latter is a parameterized model in real space.

For anomalous pairs, difference data $\Delta F^{\text{ano}}_{\mathbf{h}} = |F^{+}_{\mathbf{h}}| - |F^{-}_{\mathbf{h}}|$ can be computed using consistent Friedel-pair conventions. The package also supports phasing-related transforms, French–Wilson analysis, and time-resolved difference map workflows. It is a framework for manipulating and analyzing diffraction data, *not* an alternative electron microscope or a single monolithic structure-solution algorithm.

15 Recovering phases: four mathematically different strategies

15.1 Molecular replacement

If a homologous protein or predicted model is available, apply candidate rotations R and translations $\mathbf{t}$ within the unit cell, calculate predicted amplitudes, and optimize agreement with observations. The data cannot determine the overall origin, so translations are relative to a chosen crystallographic origin convention. A simplified molecular-replacement score could be

$$L(R, \mathbf{t}) = -\sum_{\mathbf{h}} \log p\big(|F_{\text{obs},\mathbf{h}}| \mid F_{\text{calc},\mathbf{h}}(R, \mathbf{t})\big), \tag{51}$$

with the actual likelihood accounting for model error, crystallographic symmetry and uncertainties. Once an initial model is placed, its calculated phases $\phi_{\text{calc},\mathbf{h}} = \arg F_{\text{calc},\mathbf{h}}$ can be used for electron-density synthesis. MR is model-assisted and can inherit model bias.

15.2 SAD anomalous diffraction

The complex atomic scattering factor near an absorption edge is $f(\lambda) = f_0 + f'(\lambda) + \mathrm{i}f''(\lambda)$. The imaginary anomalous part breaks exact Friedel equivalence. SAD methods use these differences, together with chemical and probabilistic information, to estimate phases or a phase probability distribution. One wavelength provides anomalous signal without requiring multiple isomorphous crystals, but its phasing power depends on anomalous scatterer content, signal quality and resolution.

15.3 Isomorphous replacement

Compare native and heavy-atom derivative crystals. In the simplest case $F_{PH} = F_P + F_H$ as complex vectors, where F_P is native protein, F_H the heavy-atom contribution and F_{PH} the derivative. Measurements give magnitudes $|F_P|, |F_{PH}|$ while modeling heavy-atom sites estimates F_H. Intersections of vector constraints determine possible phases; multiple derivatives or anomalous data help remove ambiguities. Non-isomorphism and scaling error can severely limit accuracy.

15.4 Density modification and real-space constraints

After initial phasing, constraints such as solvent flattening, histogram matching, noncrystallographic symmetry averaging and positivity-related considerations can improve phases. In broad mathematical terms, density modification alternates between Fourier-domain amplitude constraints and real-space priors. It does not evade the need for genuine information: imposing a strong wrong prior can create misleading maps.

15.5 What a phase probability distribution means

Rather than storing one certain phase $\phi_{\mathbf{h}}$, a method may represent $p(\phi_{\mathbf{h}} \mid \text{data})$. A phase expectation is

$$\mathbb{E}[e^{i\phi_{\mathbf{h}}} \mid \text{data}] = m_{\mathbf{h}} e^{i\phi_{\text{best},\mathbf{h}}}, \quad 0 \le m_{\mathbf{h}} \le 1. \tag{52}$$

The factor m is a phase figure of merit in a standard sense: uncertain, widely dispersed phases have smaller resultant length. This explains why map coefficients often include phase-information weights instead of merely adopting all phase estimates as exact.

16 From amplitudes and phases to electron-density maps

16.1 Fourier synthesis from discrete reflections

A practical finite-resolution map has the form

$$\boxed{\rho_{\text{map}}(\mathbf{x}) = \frac{1}{V_c} \sum_{\mathbf{h} \in \mathcal{H}} W_{\mathbf{h}} |F_{\mathbf{h}}| e^{i\phi_{\mathbf{h}}} e^{-2\pi i \mathbf{h}\cdot\mathbf{x}},} \tag{53}$$

where $\mathcal{H}$ is the measured reflection set, $W_{\mathbf{h}}$ is an optional complex or real weighting/filter coefficient, and the sum is real after appropriate Friedel symmetry for a real density. A map grid discretizes fractional coordinates as $x_a = a/N_x$, etc.; fast Fourier transforms evaluate the synthesis efficiently on a regular grid.

Interpretation: every reflection is a spatial sinusoid with amplitude and phase. Adding all of them reconstructs spatial density. Missing high-frequency terms blur peaks; missing reflections and abrupt truncation create artifacts.

16.2 $2mF_o - DF_c$ and $mF_o - DF_c$ maps

After refinement, a common family of likelihood-weighted map coefficients is

$$C_{2mF_o - DF_c, \mathbf{h}} = (2m_{\mathbf{h}} |F_{\text{obs},\mathbf{h}}| - D_{\mathbf{h}} |F_{\text{calc},\mathbf{h}}|) e^{i\phi_{\text{calc},\mathbf{h}}}, \tag{54}$$

$$C_{mF_o - DF_c, \mathbf{h}} = (m_{\mathbf{h}} |F_{\text{obs},\mathbf{h}}| - D_{\mathbf{h}} |F_{\text{calc},\mathbf{h}}|) e^{i\phi_{\text{calc},\mathbf{h}}}. \tag{55}$$

Here m is a phase/model-weighting term and D is a model-error attenuation factor, both estimated under a crystallographic likelihood model. The exact conventions and coefficient meanings vary across software. These names should not be read as literal equality of $m = D = 1$ in all datasets.

The $2mF_o - DF_c$ map helps inspect supported density around modeled atoms. The $mF_o - DF_c$ difference map highlights residual mismatch: a positive peak may indicate missing electron density, and a negative peak may indicate excess modeled density, but artifacts and scale/phase errors also contribute. Refined model phases cause model bias, so *no map is automatically objective simply because it uses observed amplitudes.*

17 Atomic refinement as regularized maximum-likelihood inference

17.1 The atomic parameter vector

Let $\boldsymbol{\theta} = \{\mathbf{x}_j, o_j, B_j, \text{solvent, scales}, \dots\}$ contain atomic positions, occupancies, displacement parameters and other model variables. The forward operator calculates $F_{\text{calc},\mathbf{h}}(\boldsymbol{\theta})$ by Eq. (14). The structure refinement problem is to infer $\boldsymbol{\theta}$ from experimental diffraction and chemical knowledge.

17.2 Why least squares alone overfits

A simple amplitude residual is

$$L_X(\boldsymbol{\theta}) = \sum_{\mathbf{h}} w_{\mathbf{h}} \left(|F_{\text{obs},\mathbf{h}}| - k |F_{\text{calc},\mathbf{h}}(\boldsymbol{\theta})| \right)^2. \tag{56}$$

k is an overall scale; $w_{\mathbf{h}}$ are precision weights. However, at modest resolution there are often too few independent measurements to determine every atomic coordinate and disorder parameter freely. Without restraints, atoms can move into implausible geometries while improving fit to noisy data.

17.3 A posterior with geometry restraints

Use Bayes' rule:

$$p(\boldsymbol{\theta} \mid \text{data}) \propto p(\text{data} \mid \boldsymbol{\theta}) p_{\text{chem}}(\boldsymbol{\theta}). \tag{57}$$

Taking minus logs gives the optimization target

$$\boxed{E_{\text{total}}(\boldsymbol{\theta}) = E_X(\boldsymbol{\theta}) + \lambda E_{\text{geom}}(\boldsymbol{\theta}).} \tag{58}$$

E_X is a crystallographic negative log likelihood or related target. E_{geom} penalizes deviations from expected stereochemistry: bonds, angles, torsions, chirality, planarity, nonbonded clashes and (when appropriate) reference structures or secondary structure. λ controls the balance. PHENIX implements a weighted combination of X-ray and geometry terms, with its own dynamic weight conventions [7].

An elementary bond restraint between atoms i, j is

$$E_{ij}^{\text{bond}} = \frac{1}{2}\left(\frac{\|\mathbf{r}_i - \mathbf{r}_j\| - d_{ij}^0}{\sigma_{ij}^0}\right)^2, \tag{59}$$

where d_{ij}^0 is the expected bond length and σ_{ij}^0 a restraint tolerance. This is mathematically a Gaussian prior on bond length.

17.4 Structure-factor coordinate derivative

For clarity ignore coordinate dependence of f_j and isotropic B_j. An atom contributes $F_{j\mathbf{h}} = o_j f_j e^{-B_j s^2/4} e^{2\pi i \mathbf{g}_{\mathbf{h}} \cdot \mathbf{r}_j}$. Then

$$\boxed{\frac{\partial F_{\text{calc},\mathbf{h}}}{\partial \mathbf{r}_j} = 2\pi i \mathbf{g}_{\mathbf{h}} F_{j\mathbf{h}}.} \tag{60}$$

This follows by differentiating an exponential, $\nabla_{\mathbf{r}_j} e^{2\pi i \mathbf{g}\cdot\mathbf{r}_j} = 2\pi i \mathbf{g}\, e^{2\pi i \mathbf{g}\cdot\mathbf{r}_j}$. It shows why a differentiable structure-factor calculator can connect Cartesian atom motion to diffraction residuals and neural structural representations.

17.5 R_{work} and R_{free}

A conventional amplitude residual is

$$R = \frac{\sum_{\mathbf{h}\in\mathcal{S}} \big||F_{\text{obs},\mathbf{h}}| - k|F_{\text{calc},\mathbf{h}}|\big|}{\sum_{\mathbf{h}\in\mathcal{S}} |F_{\text{obs},\mathbf{h}}|}. \tag{61}$$

R_{work} uses reflections included in refinement; R_{free} uses a held-out set excluded from parameter fitting. A low training residual alone is not evidence against overfitting; the held-out metric is intended to estimate generalization. The free set must be protected throughout model building and map-based decisions. Reflections can be statistically correlated through symmetry, NCS and other structure; the split strategy matters.

17.6 Occupancy versus disorder: a key identifiability problem

For a single atom, amplitude scales approximately as $o_j e^{-B_j s^2/4}$. Low occupancy o_j and high disorder B_j both reduce high-resolution contribution. Their effects are correlated, especially over limited resolution range. Consequently, a weak feature is not automatically proof of a low population state. Interpretation requires adequate resolution, restraints, independent evidence, and appropriately matched comparisons.

18 Model validity beyond one agreement number

A credible atomic model should pass bond/angle, Ramachandran, rotamer, clash and symmetry-contact checks; be chemically plausible; fit both ordinary and difference electron-density maps; and maintain an acceptable R_{free} relative to R_{work}. Local coordinate confidence depends on resolution, occupancy, disorder and map features. Directly comparing two coordinate sets requires common alignment, uncertainty analysis and a check that their crystallographic environments or unit-cell properties have not changed enough to invalidate naive comparison.

19 Comparative crystallography: the Hekstra-lab difference-map viewpoint

19.1 The exact difference of two complex Fourier coefficients

For two states, OFF and ON, their exact density difference is

$$\Delta\rho(\mathbf{x}) = \frac{1}{V_c}\sum_{\mathbf{h}} \left(F_{\mathbf{h}}^{\text{ON}} - F_{\mathbf{h}}^{\text{OFF}}\right) e^{-2\pi i \mathbf{h}\cdot\mathbf{x}}. \tag{62}$$

This uses *complex* differences, including both phase and amplitude changes. Experimentally, amplitudes are measured but ON/OFF phases are generally not known independently.

19.2 Practical isomorphous difference Fourier maps

For sufficiently isomorphous datasets, a widely used approximation is

$$\boxed{\Delta\rho_{\text{approx}}(\mathbf{x}) = \frac{1}{V_c}\sum_{\mathbf{h}} w_{\mathbf{h}} \left(|F_{\text{ON},\mathbf{h}}| - |F_{\text{OFF},\mathbf{h}}|\right) e^{i\phi_{\text{ref},\mathbf{h}}} e^{-2\pi i \mathbf{h}\cdot\mathbf{x}}.} \tag{63}$$

Here $\phi_{\text{ref},\mathbf{h}}$ commonly comes from an OFF-state reference model and $w_{\mathbf{h}}$ is a difference weighting. Using one reference phase set makes a sensitive comparison possible without independently refining the ON model. This is an *approximation*: the neglected phase differences may matter, especially for large changes.

Inputs: matched reflection indices, consistently scaled ON and OFF amplitudes, reference phases, uncertainties, and matching symmetry/coordinate frames. **Output:** positive and negative regions in a real-space difference density grid. In such maps a paired positive/negative feature near an atom can indicate density moving from one location to another; it is not automatically a force vector.

19.3 Why differences can be more sensitive than comparing refined coordinates

Suppose an atomic coordinate shifts by only $\delta\mathbf{r}$. Two independently refined models have coordinate errors ϵ_A, ϵ_B, so the measured difference is $\delta\mathbf{r} + (\epsilon_B - \epsilon_A)$. Its variance includes both errors and possible covariance. A difference Fourier map instead directly compares measured amplitudes before fitting a distinct perturbed-state atomic model. Shared systematic effects can cancel, making subtle collective perturbations more detectable—provided the datasets are truly comparable and carefully scaled.

19.4 Uncertainty propagation into amplitude differences

For $\Delta F = |F_{\text{ON}}| - |F_{\text{OFF}}|$,

$$\boxed{\text{Var}(\Delta F) = \sigma_{\text{ON}}^2 + \sigma_{\text{OFF}}^2 - 2\,\text{Cov}(F_{\text{ON}}, F_{\text{OFF}}).} \tag{64}$$

The covariance term matters if shared scale fitting or correlated observations induce common errors. When errors are independent, the covariance vanishes and uncertainties add in quadrature. A per-reflection significance diagnostic is $Z_{\mathbf{h}} = \Delta F_{\mathbf{h}}/\sigma_{\Delta F,\mathbf{h}}$, but a full map combines many reflections and exhibits spatial correlations.

19.5 Why a moving atom generates a positive/negative feature pair

Take a smooth one-dimensional density peak $\rho_0(x)$ displaced by a small δ. Then

$$\rho_1(x) - \rho_0(x) = \rho_0(x - \delta) - \rho_0(x) \approx -\delta \frac{\mathrm{d}\rho_0}{\mathrm{d}x}. \tag{65}$$

This Taylor expansion means the difference is approximately the spatial derivative of the original density. A Gaussian peak therefore generates a negative lobe on the side it leaves and a positive lobe on the side it enters. For 3D displacements, $\Delta\rho \approx -\delta\mathbf{r} \cdot \nabla\rho_0$. The approximation fails for large displacements, occupancy changes, and pronounced conformational mixtures.

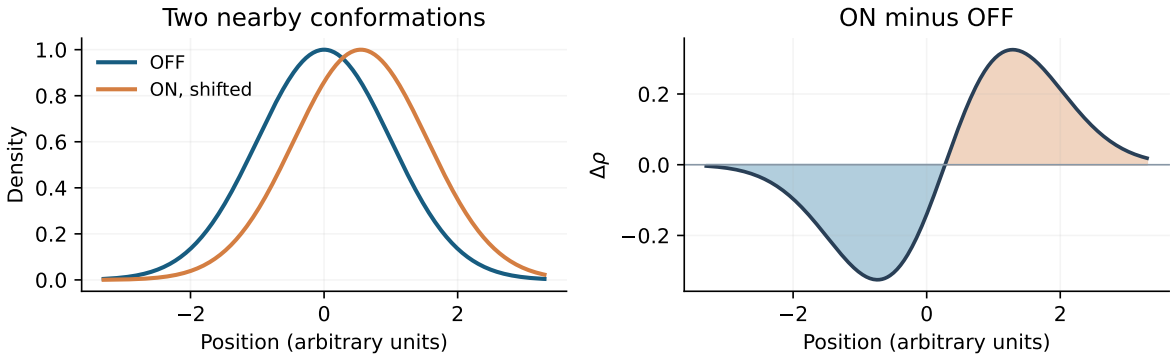


Figure 2: The difference of two nearby density peaks creates negative and positive lobes. At small displacement the difference is proportional to a spatial derivative, $\Delta\rho \simeq -\delta\,\rho_0'(x)$. This is illustrative, not an observed protein difference map.

19.6 Non-isomorphism and MatchMaps

Ordinary Eq. (63) implicitly assumes both Fourier datasets describe nearly the same periodic coordinate system. When unit-cell lengths, angles or crystal form change, directly subtracting coefficients at the same Miller indices can compare different physical spatial frequencies: $\mathbf{g}_{\mathbf{h}}^{(A)} = B_A\mathbf{h}$ differs from $\mathbf{g}_{\mathbf{h}}^{(B)} = B_B\mathbf{h}$. This can generate strong artifacts unrelated to true protein change.

MatchMaps (Brookner and Hekstra, 2024) addresses non-isomorphous comparative diffraction by placing datasets into an appropriately comparable frame while retaining the sensitivity of observed difference maps [4]. Conceptually, the required operation is closer to matching real-space maps, coordinates and support before subtraction than to assuming identical Miller labels are physically identical Fourier points. Its actual algorithm is more specific than simply subtracting maps after rigid superposition.

20 Difference-map weighting and the 2025 Double Wilson framework

20.1 Why weight amplitude differences?

Poorly measured reflections can dominate noisy Fourier differences. One common family of weights depends on each reflection's amplitude difference, uncertainty and representative dataset-wide mean scales; the precise function is chosen to downweight weakly informative or outlying values. For example, a simple uncertainty-only illustration is

$$w_{\mathbf{h}} = \frac{1}{1 + \sigma^2_{\Delta F,\mathbf{h}}/\tau^2}, \tag{66}$$

where τ is a positive scale chosen for this illustrative shrinkage model. **This equation is an example, not the specific weighting function used in EF-X or reciprocalspaceship.** Hekstra et al. (2016) and Greisman et al. (2021) discuss distinct empirical/statistical weighting families used for time-resolved maps [1, 2].

20.2 Correlated priors across multiple states

If OFF and ON datasets correspond to related structures, their true Fourier coefficients are not independent random variables. A statistical comparison can encode that correlation instead of merging states independently and taking a noisy difference. In generic form, a prior may connect state-specific latent structure factors,

$$p(F^{\mathrm{OFF}}, F^{\mathrm{ON}} \mid \boldsymbol{\eta}), \tag{67}$$

where hyperparameters $\boldsymbol{\eta}$ describe their expected similarity. A posterior then combines the paired prior with state-specific measurement likelihoods:

$$p(F^{\mathrm{OFF}}, F^{\mathrm{ON}} \mid Y^{\mathrm{OFF}}, Y^{\mathrm{ON}}) \propto p(Y^{\mathrm{OFF}} \mid F^{\mathrm{OFF}}) p(Y^{\mathrm{ON}} \mid F^{\mathrm{ON}}) p(F^{\mathrm{OFF}}, F^{\mathrm{ON}}). \tag{68}$$

In a controlled perturbation, the shared structural baseline is information: a prior that respects related states can increase sensitivity to small changes, subject to calibration against induced shrinkage/bias. Hekstra et al. (2025) developed a *Double Wilson* statistical framework for sensitive comparative crystallography, integrated with the Careless research program [5]. The equations above express the general probabilistic idea and are **not** a claim to reproduce the paper's complete specialized parameterization.

20.3 A useful caution about difference amplitudes

Do not confuse $|F_{\mathrm{ON}}| - |F_{\mathrm{OFF}}|$ with $|F_{\mathrm{ON}} - F_{\mathrm{OFF}}|$. The first is a *signed difference of magnitudes*; the second is the *magnitude of a complex difference*. Their values are generally unequal. Difference Fourier synthesis uses signed amplitude differences together with an estimated phase, not magnitudes of differences as if the latter were directly measurable.

21 EF-X: electrically stimulated and time-resolved protein crystallography

21.1 What is measured, and what is intervened on?

Electric-field-stimulated X-ray crystallography (EF-X) applies a timed electric-field pulse while collecting diffraction snapshots. The original Hekstra and colleagues demonstration used a PDZ domain to observe sub-microsecond field-stimulated structural responses [1]. It is not simply another static structure-solution experiment: the manipulated variable is the applied field waveform $\mathbf{E}(t)$, while the observations remain diffraction intensities of Bragg reflections. The data may be collected at several delays and for different field polarities.

In a classical molecular description an electric field couples to charges q_j at positions $\mathbf{r}_j$, giving the perturbation energy

$$\boxed{U_E(\mathbf{r}_1, \ldots, \mathbf{r}_M; t) = -\sum_j q_j \mathbf{E}(t) \cdot \mathbf{r}_j = -\mathbf{E}(t) \cdot \mathbf{M},} \tag{69}$$

where $\mathbf{M} = \sum_j q_j \mathbf{r}_j$ is the molecular dipole relative to a chosen reference and charge model. In a net-charged system dipole position depends on origin; physically meaningful force calculations require electric field geometry, counterions, dielectric response, solvent polarization and constraints. The elementary electrostatic force is $\mathbf{f}_j = q_j \mathbf{E}$, but treating each protein atom as a free vacuum charge would be an oversimplification.

21.2 Equilibrium response derived from statistical mechanics

Let X be a structural observable (a coordinate, domain angle, pair distance, or some linear functional of density) and $M_\parallel$ the component of dipole along the applied field. Consider a constant scalar field E with equilibrium probability

$$p_E(\Gamma) = \frac{e^{-\beta[U_0(\Gamma) - E M_\parallel(\Gamma)]}}{Z(E)}, \quad Z(E) = \int e^{-\beta[U_0 - E M_\parallel]} \mathrm{d}\Gamma, \tag{70}$$

where Γ denotes all microscopic coordinates and $\beta = (k_B T)^{-1}$. Differentiating the expectation $\langle X\rangle_E = \int X p_E \mathrm{d}\Gamma$ yields

$$\boxed{\frac{\partial\langle X\rangle_E}{\partial E} = \beta\left(\langle X M_\parallel\rangle_E - \langle X\rangle_E\langle M_\parallel\rangle_E\right) = \beta\,\mathrm{Cov}_E(X, M_\parallel).} \tag{71}$$

Derivation: $\partial_E \log Z = \beta\langle M_\parallel\rangle_E$. Hence $\partial_E p_E = \beta p_E(M_\parallel - \langle M_\parallel\rangle_E)$. Substitute into $\partial_E\langle X\rangle_E$. The identity applies to an equilibrium, conservative field coupling. Pulsed EF-X generally measures *nonequilibrium time response*; Eq. (71) is a useful limiting reference, not an automatic interpretation of delay-resolved maps.

21.3 Linear dynamical response to a pulse

For sufficiently weak perturbations, linear response theory expresses

$$\boxed{\delta\langle X(t)\rangle = \int_{-\infty}^{t} \chi_{XM}(t-t')E(t')\,\mathrm{d}t',} \tag{72}$$

where $\chi_{XM}(\tau)$ is a causal response kernel ($\chi = 0$ for $\tau < 0$). A step field $E(t) = E_0\Theta(t)$ produces $\delta\langle X(t)\rangle = E_0\int_0^t \chi_{XM}(\tau)\mathrm{d}\tau$. For a simple overdamped coordinate q in a harmonic well,

$$\gamma\dot{q} + k(q - q_0) = f_0\Theta(t), \tag{73}$$

where γ is a friction coefficient, k an effective stiffness and f_0 the generalized force. Solving the first-order ODE with $q(0) = q_0$ yields

$$\boxed{q(t) - q_0 = \frac{f_0}{k}\left(1 - e^{-t/\tau}\right), \quad \tau = \gamma/k.} \tag{74}$$

Proof: subtract the stationary solution $q_\infty = q_0 + f_0/k$, giving $\gamma\dot{\delta q} + k\delta q = 0$; integrate $\mathrm{d}\delta q/\delta q = -(k/\gamma)\mathrm{d}t$ to obtain $\delta q(t) = \delta q(0)e^{-t/\tau}$. Several coupled modes produce multi-exponential or more general responses. In real data, the observable also convolves with field timing, X-ray pulse duration and sample heterogeneity.

21.4 Field reversal is an experimental symmetry test

Suppose a measured map signal $S(E)$ has a Taylor expansion around zero field:

$$S(E) = S(0) + a_1 E + a_2 E^2 + a_3 E^3 + \cdots. \tag{75}$$

Define the odd-in-field and even-in-field components

$$S_{\mathrm{odd}}(E) = \frac{S(+E) - S(-E)}{2}, \qquad S_{\mathrm{even}}(E) = \frac{S(+E) + S(-E)}{2} - S(0). \tag{76}$$

Then $S_{\mathrm{odd}} = a_1 E + a_3 E^3 + \cdots$ and $S_{\mathrm{even}} = a_2 E^2 + \cdots$. This decomposition is a symmetry diagnostic, *not* by itself proof that odd signals correspond exclusively to direct electrostatic forces or even signals exclusively to heating; multiple microscopic mechanisms can share these parities.

21.5 What the images become in EF-X

For condition c and time delay t, the pipeline yields reflection amplitudes $|F_{\mathbf{h}}(c,t)|$ or their posterior distributions. A reference structure supplies $\phi_{\mathbf{h}}^{\mathrm{ref}}$ and a time-dependent difference map is

$$\Delta\rho(\mathbf{x}; c, t) \approx \frac{1}{V_c}\sum_{\mathbf{h}} w_{\mathbf{h}ct}\left(|F_{\mathbf{h}}(c,t)| - |F_{\mathbf{h}}(\mathrm{OFF})|\right) e^{\mathrm{i}\phi_{\mathbf{h}}^{\mathrm{ref}}} e^{-2\pi\mathrm{i}\mathbf{h}\cdot\mathbf{x}}. \tag{77}$$

This yields an experimentally constrained *time series of density differences*, not an atom-resolved trajectory unless an additional dynamic structural model is fitted and validated.

22 Small shifts, occupancies, and multiple conformers: what a map can identify

22.1 Population mixtures are linear in amplitudes before taking magnitudes

Suppose the unit cell is occupied by state A with probability $1-p$ and state B with probability p, and the Bragg signal reflects coherent scattering from the average periodic density (under the relevant disorder assumptions). Then the mean structure factor is

$$\boxed{\overline{F}_{\mathbf{h}} = (1-p)F_{A,\mathbf{h}} + pF_{B,\mathbf{h}} = F_{A,\mathbf{h}} + p\Delta F_{\mathbf{h}}.} \tag{78}$$

The corresponding ideal Bragg intensity is

$$|\overline{F}_{\mathbf{h}}|^2 = |F_A|^2 + 2p\,\mathrm{Re}(F_A^*\Delta F) + p^2|\Delta F|^2. \tag{79}$$

This is *not* generally equal to $(1-p)|F_A|^2 + p|F_B|^2$. The latter is an incoherent intensity mixture and corresponds to a different physical averaging regime. Coherent Bragg peaks and diffuse scattering can partition information differently. This distinction is essential before inferring a state occupancy from a measured amplitude difference.

22.2 First-order structure-factor sensitivity to atom shifts

From Eq. (60), for small atomic displacements $\delta\mathbf{r}_j$,

$$\boxed{\delta F_{\mathbf{h}} \approx 2\pi\mathrm{i}\sum_j (\mathbf{g}_{\mathbf{h}}\cdot\delta\mathbf{r}_j)F_{j\mathbf{h}}.} \tag{80}$$

This is the multivariate first-order Taylor expansion of the forward diffraction model. It says each reflection is sensitive to a different projection of each atom's displacement along reciprocal vector $\mathbf{g}_{\mathbf{h}}$. If several atoms move coherently, their complex contributions interfere.

22.3 But amplitude sensitivity loses information

For $F = |F|e^{\mathrm{i}\phi}$ and a small perturbation δF,

$$\boxed{\delta|F| \approx \mathrm{Re}(e^{-\mathrm{i}\phi}\delta F),} \qquad \delta I = \delta|F|^2 \approx 2\,\mathrm{Re}(F^*\delta F). \tag{81}$$

Derivation: $|F+\delta F|^2 = |F|^2 + 2\,\mathrm{Re}(F^*\delta F) + O(|\delta F|^2)$. Divide the first-order intensity change by $2|F|$ when $|F| \neq 0$. This demonstrates why phase information remains important even for apparently simple structural displacements.

22.4 Spatially structured deformation versus local interaction

An observed difference near a DNA attachment site can reflect local electrostatics, steric interactions, hydration, changed occupancy, or genuine mechanical deformation; diffraction does not label the causal mechanism. For the PP2A DNA-spring constructs, matched genotypes and attachment chemistry in relaxed versus loaded conditions are required to support mechanical causality. The 2026 ubiquitin–DNA chimera preprint from the Hekstra/Gardner collaboration provides a relevant warning: attached DNA itself can alter conformational equilibria, even when duplex-induced mechanical effects are modest [9].

23 Mechanics from structural changes: identifiability, covariance, and modes

23.1 A linear mechanical model of a protein

Let $\mathbf{q} \in \mathbb{R}^d$ collect coarse-grained deformations (e.g., 15 HEAT-repeat hinge angles and translations). Near an equilibrium $\mathbf{q}_0$, approximate the free energy as

$$G_0(\mathbf{q}) \approx G_0(\mathbf{q}_0) + \tfrac{1}{2}(\mathbf{q}-\mathbf{q}_0)^{\mathsf{T}} K (\mathbf{q}-\mathbf{q}_0), \tag{82}$$

where K is a symmetric positive-definite stiffness/Hessian matrix on the nontrivial constrained coordinates. Generalized external forces $\mathbf{f}$ couple as $-\mathbf{f}^{\mathsf{T}}(\mathbf{q}-\mathbf{q}_0)$. Minimizing total free energy gives

$$\boxed{\delta\mathbf{q} = K^{-1}\mathbf{f}.} \tag{83}$$

The inverse stiffness K^{-1} is the compliance or susceptibility matrix. If the system is harmonic at thermal equilibrium,

$$\boxed{C_q = \mathbb{E}[\delta\mathbf{q}\delta\mathbf{q}^{\mathsf{T}}] = k_B T\, K^{-1}.} \tag{84}$$

This Gaussian covariance relation follows by evaluating the Boltzmann distribution for a quadratic energy. Real proteins are nonharmonic, may have multiple basins, and are constrained by lattice contacts. Crystal-state Hessians need not equal solution-state Hessians.

23.2 Diffraction measures only some combinations of the modes

Linearize the observable vector of measured amplitude differences around a reference model:

$$\mathbf{y} \approx H\,\delta\mathbf{q} + \boldsymbol{\epsilon}, \tag{85}$$

where $\mathbf{y} \in \mathbb{R}^n$ is the vector of reflection amplitude differences, $H \in \mathbb{R}^{n\times d}$ is the sensitivity Jacobian $H_{aj} = \partial|F_{\mathbf{h}_a}|/\partial q_j$, and $\boldsymbol{\epsilon}$ is observation error. If H has a nontrivial null space, vectors $\mathbf{v}$ satisfying $H\mathbf{v} = 0$ cannot be detected to first order from these measurements.

23.3 Identifiability via singular-value decomposition

Write $H = U\Lambda V^{\mathsf{T}}$ with singular values $\lambda_1 \geq \cdots \geq 0$. Small singular values correspond to combinations of deformations that weakly change observations. With Gaussian noise covariance C_ϵ and a Gaussian mechanical prior of covariance C_0, the posterior covariance is

$$\boxed{C_{\text{post}} = (H^{\mathsf{T}} C_\epsilon^{-1} H + C_0^{-1})^{-1}.} \tag{86}$$

The first term supplies data information; the second supplies prior information. A seemingly precise force estimate can be *prior dominated* when the measured reflections do not constrain the relevant deformation mode.

23.4 Force inversion requires a stiffness or calibration

Suppose structural reconstruction estimates $\delta\mathbf{q}$. To infer force by $\mathbf{f} = K\delta\mathbf{q}$, one must know K and the physical loading geometry. If the DNA spring modifies the total potential, the correct equilibrium condition is instead

$$\nabla_{\mathbf{q}} G_0(\mathbf{q}) + \nabla_{\mathbf{q}} U_{\text{DNA}}(\mathbf{q}) = 0. \tag{87}$$

The effective generalized load is $\mathbf{f}_{\text{DNA}} = -\nabla_{\mathbf{q}} U_{\text{DNA}}$, not an independently fixed scalar force across all molecules. Diffraction-derived atomic changes by themselves therefore do *not* determine exact piconewton forces.

24 Diffuse scattering and correlated motion (advanced)

24.1 From Bragg averages to pairwise correlations

Let $F_n(\mathbf{q})$ denote the fluctuating scattering amplitude of unit cell n and $\mathbf{R}_n$ its lattice position. Total amplitude is $A(\mathbf{q}) = \sum_n F_n(\mathbf{q}) e^{2\pi i \mathbf{q}\cdot\mathbf{R}_n}$. The ensemble-averaged intensity is

$$\mathbb{E}[|A(\mathbf{q})|^2] = \sum_{n,m} e^{2\pi i \mathbf{q}\cdot(\mathbf{R}_n - \mathbf{R}_m)} \mathbb{E}[F_n(\mathbf{q}) F_m(\mathbf{q})^*] \tag{88}$$

$$= |\mathbb{E}[F]|^2 \left| \sum_n e^{2\pi i \mathbf{q}\cdot\mathbf{R}_n} \right|^2 + \sum_{n,m} e^{2\pi i \mathbf{q}\cdot(\mathbf{R}_n - \mathbf{R}_m)} \operatorname{Cov}(F_n, F_m^*). \tag{89}$$

The first term is coherent Bragg scattering from mean amplitude; the second describes correlations of fluctuations and contributes diffuse scattering. $\operatorname{Cov}(F_n, F_m^*) = \mathbb{E}[(F_n - \mathbb{E}[F_n])(F_m - \mathbb{E}[F_m])^*]$. The equality assumes equivalent mean unit cells; more general cases require cell-specific means.

This suggests a complementary route to mechanical correlations. However, separating diffuse scattering from background, disorder, lattice vibrations and instrumental artifacts is challenging. Bragg-based difference maps and diffuse scattering therefore answer related but distinct questions about protein dynamics.

25 A complete downstream-analysis example for a DNA-spring experiment

Assume we obtain two crystallographic conditions on the *same* engineered PP2A A-subunit chimera: relaxed and mechanically active duplex spring. The following is a worked *analysis design*, not a claim that these datasets already exist or will necessarily crystallize.

1. **Image acquisition and reduction.** Record detector arrays $D_j[p]$ and metadata for each condition. Calibrate detector geometry; index spots via Eq. (20); integrate reflection intensities via Eq. (22) or profile fitting Eq. (25). Keep raw integration estimates, including negative weak observations, with standard uncertainties.

2. **Scale and merge independently, then match.** Infer per-observation scales and reflection amplitudes while validating $\mathrm{CC}_{1/2}$, completeness, multiplicity, anisotropy and held-out predictions. Careless or classical scaling software can be used, provided conventions and cross-validation are explicit. Use matched resolution ranges and assess unit-cell differences.

3. **Establish a reliable reference phase model.** Use molecular replacement or an appropriate experimental phasing method. Refine an OFF/relaxed model with chemical restraints and evaluate R_{work}, R_{free} and local map quality. Retain its phase estimates $\phi_{\mathrm{ref},\mathbf{h}}$.

4. **Compute an observed difference map.** If sufficiently isomorphous, synthesize Eq. (63); if not, use a validated non-isomorphous procedure such as MatchMaps. Check peaks under alternative weighting and scaling choices, symmetry-equivalent copies, and independent crystal replicates.

5. **Test whether difference density encodes displacement.** Look for spatially coherent positive/negative density features and compare with the derivative model Eq. (65). Do not assume that every peak pair is a direct atom trajectory. Where supported by data, refine alternative conformers or restrained deformation models, with occupancy and B-factor sensitivity tests.

6. **Connect differences to mechanics.** Compute structural observables: attachment-site distance, HEAT-repeat relative rotations, inter-repeat displacement and local map correlation. Use a calibrated DNA-spring potential and independent solution-state controls to test force-response hypotheses. Evaluate whether data constrain those modes by sensitivity and uncertainty analysis (Eqs. (85)–(86)).

7. **Assess causality.** A separate, matched control that retains DNA chemistry but relieves elastic load helps distinguish intended mechanical effects from local DNA-protein interactions. Verify that different unit cells, contacts, irradiation damage, pH, ionic conditions and occupancy do not explain the structural differences.

Biological interpretation for PP2A. The A subunit is a flexible 15-HEAT-repeat scaffold. In the Hekstra-lab setting, the strongest crystallographic result would not merely be two refined coordinate models with different RMSD values; it would be reproducible, uncertainty-aware observed difference density linked to a specifically controlled mechanical perturbation, followed by conservative structural and functional interpretation.

26 X-ray crystallography versus single-particle cryo-EM: mathematical comparison

	X-ray crystallography	Single-particle cryo-EM
Primary measurement	Bragg spot intensities $\propto \lvert F_{\mathbf{h}}\rvert^2$	Noisy real-space image pixels of 2D projections.
Fourier information	Reciprocal-lattice sampled magnitudes; phases missing.	Fourier central planes, modified by CTF; poses unknown.
Main geometric hidden variables	Crystal orientation, unit cell, lattice symmetry.	Per-particle 3D orientation and 2D translation.
Structural output	Periodic unit-cell electron density, atomic models.	3D electrostatic potential map, atomic models.
Variation	Difference maps, multistate refinement, diffuse scattering, time-resolved datasets.	3D classification, 3DVA, 3DFlex, cryoDRGN and related methods.
Major inferential obstacle	Phase retrieval, scale/model bias, disorder and crystal contacts.	Severe particle SNR, pose–heterogeneity ambiguity, preferred orientation.
Mechanical inference	Compare matched perturbation states (EF-X, DNA springs) with appropriate crystal controls.	Infer condition-dependent conformational populations with calibrated loading assumptions.

27 Mathematical conclusions and limitations

1. **The native unknown is a complex Fourier field.** The electron density is reconstructed from complex coefficients, while conventional diffraction records primarily their squared magnitudes. The phase problem is central, not a peripheral technicality.
2. **Raw measurements must be reduced statistically.** Indexing assigns reciprocal labels, integration estimates spot intensities with uncertainty, scaling corrects systematic variation, and merging estimates common reflection strengths.
3. **Atomic coordinates are inferred parameters.** Refinement needs diffraction likelihoods and stereochemical priors; a good numerical residual alone does not imply physically correct coordinates, occupancies or dynamics.
4. **Difference density is often a more direct perturbation observable than the difference of separately refined structures.** But it depends on phases, correct scaling, comparability of reciprocal spaces and sound uncertainty treatment.
5. **Mechanical force is a further inverse problem.** A structural shift constrains effective deformation, not uniquely the external forces that produced it. Calibrations, controlled perturbations, physically justified energy models and uncertainty estimates are necessary.
6. **The Hekstra-lab software ecosystem attacks different layers of the inverse problem.** ReciprocalSpaceship helps represent and manipulate diffraction data; Careless performs Bayesian scaling/merging; MatchMaps addresses challenging comparative maps; SFCalculator provides differentiable model-to-reflection predictions. None alone replaces experimental phasing, model refinement and scientific validation.

A Practical vocabulary: files and data structures

Object or file	Mathematical content	Typical purpose
Image arrays (CBF/HDF5/NXmx, etc.)	Pixel intensities $D_j[p]$ and metadata	Integrate diffraction images.
Indexed integration file	Rows $(h, k, l, j, Y, \sigma, \mathbf{m})$	Scaling, merging, analysis.
MTZ reflection file	Typed columns $I, \sigma_I, F, \sigma_F, \phi, \text{FREE}, \ldots$	Maps, phasing, refinement.
PDB/mmCIF atomic model	Atoms with coordinates, occupancies, B factors, chemistry	Model-to-data calculations.
CCP4/MRC map	Sampled 3D density values with grid and origin	Inspect electron density.
Difference-map grid	$\Delta\rho(\mathbf{x})$ sampled in real space	Detect spatial change.

The exact formats depend on the instrument and processing program. Reflection-table column labels alone do not determine whether values are raw, scaled, merged or model-based; always inspect metadata and provenance.

B A checklist for interpreting a novel mechanochemical effect

Before asserting a spring-induced force-response mechanism, establish each claim separately:

1. A specific chemically correct protein–DNA species is present (including its attachment topology and stoichiometry).
2. The comparison uses sufficiently matched biological and crystallographic conditions.
3. Diffraction differences are statistically reproducible across independent data collections or crystals.
4. Difference features survive reasonable choices of scaling, merging, difference weights and reference phases.
5. The inferred geometric change is identifiable at the available resolution and supported by density.
6. Local DNA contact effects and crystal packing are distinguished from the intended elastic perturbation.
7. Numerical force estimates come from independently justified mechanical models and include uncertainty.
8. Functional conclusions rely on matching biochemical measurements rather than assuming structure alone proves catalytic regulation.

C Selected reading and source provenance

The references below include Hekstra-lab research, relevant software and foundational refinement documentation. They support the lab-specific descriptions; the introductory scattering and inverse-problem derivations are standard mathematical developments. The document uses simplified models where explicitly stated, not exact reverse-engineering of every software implementation.

References

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End of mathematical exposition